

UNION-INTERSECTION TESTS FOR SPHERICITY

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Summary: Two union-intersection tests for the sphericity hypothesis are derived using the technique of Olkin and Tomsy (1981). An extension of the Kantorovich inequality is given in terms of elementary symmetric functions.

1. INTRODUCTION

A new class of multivariate tests based on the union-intersection principle has been developed by Olkin and Tomsy (1981). These tests are given as elementary symmetric functions of certain characteristic roots. We shall use this method to derive two classes of test statistics for the sphericity hypothesis

$$(1.1) \quad H: \Sigma = \sigma^2 I_p, \text{ with } \sigma^2 \text{ unknown, against } A: \Sigma \neq \sigma^2 I_p.$$

See Anderson (1958) for a detailed exposition of this hypothesis. The union-intersection test statistics derived by Venables (1976)

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for (1.1), follow as special cases of the two new classes of test statistics.

The test statistic derived in section 2 is obtained by using the inequalities between adapted elementary symmetric functions given by Maclaurin (1729). The test statistic in section 3 is obtained by means of the extremal result given in section 4 which is a generalization of the Kantorovich inequality extension given by Khatri and Rao (1981). The formal proof of the extremal result is given as a theorem in section 5.

The notation of Olkin and Tomskey (1981), $E_j[\lambda_1, \dots, \lambda_p]$ for a class of elementary symmetric functions is used in this paper where $\lambda_1, \dots, \lambda_p$ are the characteristic roots of a $p \times p$ symmetric matrix A .

Also

$$\text{tr}_j(A) \equiv E_j[\lambda_1, \dots, \lambda_p]$$

2. ADAPTED ELEMENTARY SYMMETRIC FUNCTIONS AS UNION-INTERSECTION TEST STATISTICS

Venables (1976) decomposed (1.1) into

$$(2.1) \quad H = \bigcap_{K \in O(p)} H^*(K) \quad \text{and} \quad A = \bigcup_{K \in O(p)} A^*(K),$$

where

$H^*(K)$: Every k_j is an eigenvector of Σ ,

$A^*(K)$: $H^*(K)$ is possibly false,

and $K = [k_1, \dots, k_p]$ is a given orthogonal matrix. Let $\lambda_1, \dots, \lambda_p$ denote the characteristic roots of the $p \times p$ sample covariance matrix S .

Using the likelihood ratio statistic (Anderson (1958), Sect. 9.2)

$$(2.2) \quad |S| / \prod_{i=1}^p k_i S_{k_i},$$

for testing the component hypothesis $H^*(K)$, Venables proved that the union-intersection test statistic for (2.1) is the likelihood ratio statistic

$$(2.3) \quad T_p = \prod_{i=1}^p \ell_i / (\sum_{i=1}^p \ell_i / p)^p.$$

The fundamental inequality between the arithmetic and geometric means follows as a special case of the inequalities between adapted elementary symmetric functions. From Maclaurin (1729) we have that

$$(2.4) \quad \tilde{E}_p^{1/p} \leq \tilde{E}_{p-1}^{1/(p-1)} \leq \dots \leq \tilde{E}_2^{1/2} \leq \tilde{E}_1,$$

where $\tilde{E}_j$ denotes the j -th adapted elementary symmetric function and $\tilde{E}_j = E_j[\theta_1, \dots, \theta_p] / \binom{p}{j}$, $j \leq p$ and $\theta_1, \dots, \theta_p$ are the characteristic roots of Σ .

Anderson (1958) used the inequality between $\tilde{E}_p^{1/p}$ and $\tilde{E}_1$ to derive the likelihood ratio test statistic (2.3). Thy hypothesis H is true if and only if all the roots of $|\Sigma - \theta I_p| = 0$ are equal or equivalently that $\tilde{E}_p^{1/p} = \tilde{E}_1$. This algebraic argument will also hold for the intermediate functions of (2.4) if

$$(2.5) \quad \tilde{E}_p / \tilde{E}_1^p = \tilde{E}_{p-1} / \tilde{E}_1^{p-1} = \dots = \tilde{E}_2 / \tilde{E}_1^2 = 1.$$

Now proceeding with the method of Olkin and Tomskey (1981) we have from (2.2) and (2.4) that a likelihood ratio component test matrix for testing $H^*(K)$ against $A^*(K)$ is

$$C_m = (K'SK)^{(p)}_j^{-1} / \left(\prod_{i=1}^p \underline{k}_i' S \underline{k}_i \right)^{j/p}$$

and the component hypothesis $H^*(K)$ is rejected against $A^*(K)$ if $\text{tr}_j(C_m) < c$. The union-intersection test statistic for (2.1) will therefore be

$$T_j = \min_{K \in O(p)} \text{tr}_j(C_m).$$

Let $L_j = \text{tr}_j(S) / \binom{p}{j}$. From (2.4) we have that

$$\left(\prod_{i=1}^p \underline{k}_i' S \underline{k}_i \right)^{j/p} \leq \left(\sum_{i=1}^p \underline{k}_i' S \underline{k}_i / p \right)^j = L_1^j$$

and therefore $\text{tr}_j(C_m) \geq L_j / L_1^j$. Hence we can say that $T_j \geq L_j / L_1^j$ and Venables (1976) showed that this overall minimum can be achieved by finding an orthogonal matrix K such that $K'SK$ has equal diagonal elements. The union-intersection test statistic

$$T_j = (E_j[\ell_1, \dots, \ell_p] / \binom{p}{j}) / (E_1[\ell_1, \dots, \ell_p] / p)^j, \quad 1 < j \leq p$$

corresponds to (2.5) and T_p is equal to the likelihood ratio statistic (2.3).

3. A CLASS OF UNION-INTERSECTION TEST STATISTICS

Consider an orthogonal matrix $K = (K_1, K_2)$, where $K_1: p \times k$, $K_2: p \times p-k$ and $k \leq p/2$. Mallows (1961) noted that testing (1.1) is equivalent to testing the independence between $\underline{x}_1 = K_1' \underline{x}$ and $\underline{x}_2 = K_2' \underline{x}$, where $\underline{x}(p \times 1)$ is distributed $N(\underline{\mu}, \Sigma)$. The decomposition of (1.1) is therefore

$$(3.1) \quad H = \bigcup_{K \in O(p)} H_{K_1 K_2} \quad \text{against} \quad A = \bigcup_{K \in O(p)} A_{K_1 K_2},$$

where $H_{K_1 K_2} : K_1' \Sigma K_2 = 0$ and $A_{K_1 K_2} : K_1' \Sigma K_2 \neq 0$.

From Venables (1976) and Srivastava and Khatri (1979) we consider the likelihood ratio component test matrix as

$$C_m = K_1' S K_1 (K_1' S^{-1} K_1)^{-1}.$$

The component hypothesis $H_{K_1 K_2}$ is rejected against $A_{K_1 K_2}$ if $\text{tr}_j(C_m) > c$. The union-intersection test statistic for (3.1) is therefore

$$T_{jk} = \max_{K_1' K_1 = I_k} \text{tr}_j(C_m), \quad j \leq k.$$

Using the upper bound given in (4.1) we have that

$$(3.2) \quad T_{jk} = E_j[\delta_1, \dots, \delta_k], \quad j \leq k,$$

where $\delta_i = (\ell_i + \ell_{p-i+1})^2/4 \ell_i \ell_{p-i+1}$ and $k=1, 2, \dots, [p/2]$, $[t]$ equals the largest integer contained in t . Furthermore $\ell_1 > \dots > \ell_p > 0$ are the ordered characteristic roots of S .

The test statistic derived by Venables (1976) is the special case $j=k$, that is

$$T_{kk} = \prod_{i=1}^k (\ell_i + \ell_{p-i+1})^2/4 \ell_i \ell_{p-i+1}.$$

4. AN EXTENSION OF THE KANTOROVICH INEQUALITY

A natural multivariate extension of the Kantorovich inequality is given by the comparison of the positive definite matrices $X'AX$ and $(X'A^{-1}X)^{-1}$ where X is an $n \times k$ matrix of rank k such that $X'X = I_k$ and A is a symmetric $n \times n$ matrix with

characteristic roots $\lambda_1 \geq \dots \geq \lambda_n > 0$ (Khatri and Rao (1981)). The j -th elementary symmetric function of the characteristic roots of $X'AX$ with respect to $(X'A^{-1}X)^{-1}$ is used as a measure of comparison between the two matrices. We obtain the following upper bounds for this function,

$$(4.1) \quad \text{tr}_j(X'AXX'A^{-1}X) \leq E_j[\delta_1, \dots, \delta_k] \quad \text{if } 2k \leq n$$

$$\leq E_j[\delta_1, \dots, \delta_{n-k}, 1, \dots, 1] \quad \text{if } 2k > n,$$

where $\delta_m = (\lambda_m + \lambda_{n-m+1})^2/4 \lambda_m \lambda_{n-m+1}$ and there are $2k-n$ ones for the case $2k > n$.

Several known inequalities follow from (4.1) for certain combinations of j and k . When $j=k$ we have the results of Bloomfield and Watson (1975) and Knott (1975) that

$$|X'AX| |X'A^{-1}X| \leq \prod_{i=1}^{\min(k, n-k)} \delta_i.$$

The result of Khatri and Rao (1981) follows when $j=1$ in (4.1), that is

$$\text{tr}(X'AXX'A^{-1}X) \begin{cases} \leq \sum_{i=1}^k \delta_i & \text{if } 2k \leq n \\ \leq \sum_{i=1}^{n-k} \delta_i + 2k - n & \text{if } 2k > n. \end{cases}$$

If $j=1$ and $k=1$, (4.1) reduces to the Kantorovich inequality

$$(\underline{x}'A\underline{x})(\underline{x}'A^{-1}\underline{x}) \leq (\lambda_1 + \lambda_n)^2/4 \lambda_1 \lambda_n$$

where $\underline{x}(n \times 1)$ is any vector such that $\underline{x}'\underline{x} = 1$. The Kantorovich inequality is a well-known tool in the convergence analysis of unconstrained minimization problems (Luenberger (1973)).

The union-intersection test statistic (3.2) has thus been derived using the generalized Kantorovich inequality given in (4.1). In section 5 we obtain a slightly more general result than given in (4.1).

5. SOME RESULTS ON MATRIX EXTREMAL PROBLEMS

The following lemma will be used in the main theorem.

Lemma

Let Δ_1 and Δ_2 be two $n \times n$ diagonal matrices where

$$(5.1) \quad \begin{aligned} \Delta_1 &= \text{diag}(\alpha_1, \dots, \alpha_k, 0, \dots, 0), \\ \Delta_2 &= \text{diag}(\beta_1, \dots, \beta_k, 0, \dots, 0), \end{aligned}$$

with $\alpha_1 \geq \dots \geq \alpha_k > 0$ and $\beta_1 \geq \dots \geq \beta_k > 0$. If Γ_1 and Γ_2 are any two orthogonal matrices, both of order $n \times n$, then for $m = 1, \dots, k$

$$(5.2) \quad \text{tr}_m(\Gamma_1 \Delta_1 \Gamma_2 \Delta_2) \leq \text{tr}_m(\Delta_1 \Delta_2).$$

Proof

From the definition and properties of compound matrices given by Olkin and Tomskey (1981) we have that

$$(5.3) \quad \text{tr}_m(\Gamma_1 \Delta_1 \Gamma_2 \Delta_2) = \text{tr}(\Gamma_1 \Delta_1 \Gamma_2 \Delta_2)^{(m)} = \text{tr}(\Gamma_1^{(m)} \Delta_1^{(m)} \Gamma_2^{(m)} \Delta_2^{(m)}).$$

Furthermore $\Gamma_1^{(m)}$ and $\Gamma_2^{(m)}$ are each orthogonal. A theorem of von Neumann (1937) states that if Ω_1 and Ω_2 are orthogonal matrices then

$$(5.4) \quad \text{tr}(\Omega_1 \Delta_1 \Omega_2 \Delta_2) \leq \text{tr}(\Delta_1 \Delta_2)$$

and (5.4) holds with equality when

$$\Omega_1 = \begin{bmatrix} I_k & 0 \\ 0 & G_1 \end{bmatrix} \text{ and } \Omega_2 = \begin{bmatrix} I_k & 0 \\ 0 & G_2 \end{bmatrix},$$

where G_1 and G_2 are orthogonal matrices. Applying (5.4) to (5.3) we have that

$$\text{tr}(\Gamma_1^{(m)} \Delta_1^{(m)} \Gamma_2^{(m)} \Delta_2^{(m)}) \leq \text{tr}(\Delta_1^{(m)} \Delta_2^{(m)}) = \text{tr}_m(\Delta_1 \Delta_2),$$

and this proves (5.2).

The main theorem which follows is an expansion of the results of Khatri and Rao (1981), Bloomfield and Watson (1975) and Knott (1975).

Theorem

Let X and Y be $n \times k$ matrices of rank k such that $X'X = Y'Y = I_k$ and let A be any fixed nonsingular matrix with singular values $\lambda_1 \geq \dots \geq \lambda_n > 0$. Then for $j=1, \dots, k$,

$$\begin{aligned} \text{tr}_j(X' A Y Y' A^{-1} X) &\leq E_j[\delta_1, \dots, \delta_k] && \text{if } 2k \leq n, \\ &\leq E_j[\delta_1, \dots, \delta_{n-k}, 1, \dots, 1] && \text{if } 2k > n, \end{aligned}$$

where $\delta_m = (\lambda_m + \lambda_{n-m+1})^2/4 \lambda_m \lambda_{n-m+1}$ and there are $2k-n$ ones for the case $2k > n$.

Proof

The problem remains the same if A and A^{-1} are substituted with D and D^{-1} , where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$.

Let

$$D^{-\frac{1}{2}}XX'D^{\frac{1}{2}} = P\Delta_1Q' \quad \text{and} \quad D^{-\frac{1}{2}}YY'D^{\frac{1}{2}} = R\Delta_2S'$$

be singular value decompositions where Δ_1 and Δ_2 are given in (5.1) and P, Q, R, S are orthogonal matrices. From (5.2) we have that

$$\text{tr}_j(X'DYY'D^{-1}X) = \text{tr}_j(\Delta_1Q'S\Delta_2R'P) \leq \text{tr}_j(\Delta_1\Delta_2).$$

Furthermore $\text{tr}_j(\Delta_1\Delta_2) \leq (\text{tr}_j(\Delta_1^2))^{\frac{1}{2}}(\text{tr}_j(\Delta_2^2))^{\frac{1}{2}}$ and equality is attained when $\Delta_1 = \Delta_2$. Therefore the maximization of $\text{tr}_j(X'DYY'D^{-1}Y)$ can be restricted to the case $X=Y$. Defining $T = X'DXX'D^{-1}X$, we minimize the expression

$$\phi = \text{tr}_j(T) - \text{tr}_j(\Lambda_1 X'X)$$

where Λ_1 is the Lagrangian multiplier. Using the matrix derivative of the j -th elementary symmetric function for a mathematically independent and variable matrix discussed in Nel (1980), we have that

$$\begin{aligned} \frac{\partial \phi}{\partial X} = & (-1)^{j-1} \sum_{i=0}^{j-1} (-1)^i [\text{tr}_i(T)(D^{-1}XX'DX(T')^{j-i-1} + D^{-1}XT^{j-i-1}X'DX + \\ & DX(T')^{j-i-1}X'D^{-1}X + DXX'D^{-1}XT^{j-i-1}) - \text{tr}_i(X'X\Lambda_1)(X\Lambda_1(X'X\Lambda_1)^{j-i-1} + \\ & X(\Lambda_1X'X)^{j-i-1}\Lambda_1)] = 0. \end{aligned}$$

The equation for X after eliminating the Lagrangian multiplier is

$$(5.5) \quad \sum_{i=0}^{j-1} X(T^{j-i} + T^{j-i}) = \sum_{i=0}^{j-1} (D^{-1}X(X'DX)T^{j-i-1} + DX(X'D^{-1}X)T^{j-i-1}).$$

Khatri and Rao (1981) proved for $j=1$ in (5.5) that $P'TP$ is symmetric or T is symmetric, where P is an orthogonal matrix

and therefore T^{j-i} will also be symmetric. Postmultiplying both sides of (5.5) with $(T^{-1})^{j-i}$ we have that

$$2X = D^{-1}X(X'D^{-1}X)^{-1} + DX(X'DX)^{-1}$$

which is equivalent to the equation obtained by Bloomfield and Watson (1975) and Knott (1975). Since T is of rank k , the upper bound for $\text{tr}_j(T)$, when $2k > n$, is given by $E_j[\delta_1, \dots, \delta_{n-k}, 1, \dots, 1]$, where there are $2k-n$ ones.

Hence the theorem is proved.

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